

Computer Organisation

Topic 1 Introduction (Part 2)

Outline

- History of computers
- X86, PowerPC and ARM evolution

Organisation
and
architecture

Structure
and function

Computer
evolution and
performance

Cloud
Computing

Performance
Issues

What mobile phone are you using?



Image source: https://commons.wikimedia.org/wiki/File:Mobile_phone_timeline.png

What mobile phone are you using?

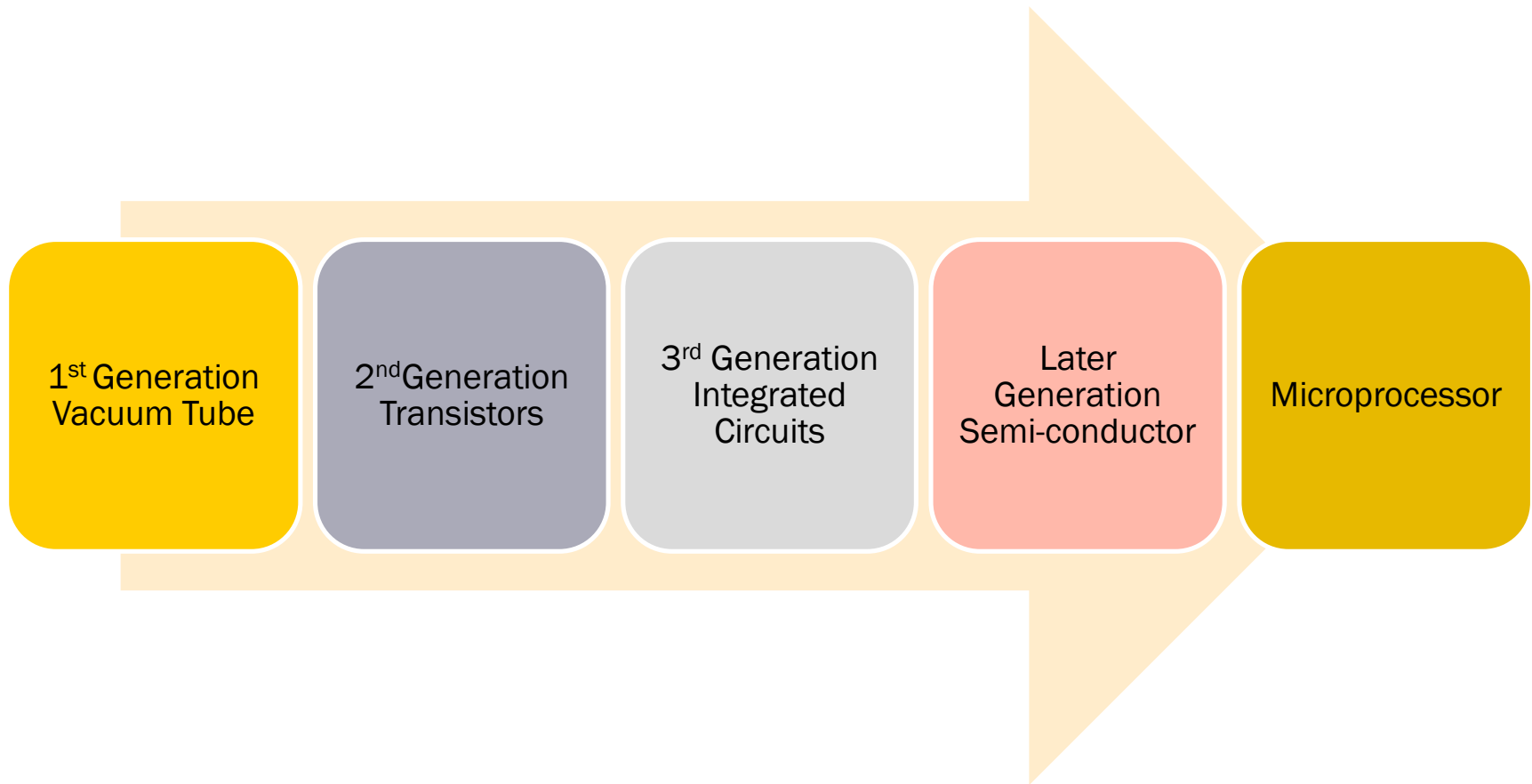


What **processor** does it use?



Image source: <https://pixabay.com/photos/cellphones-smartphones-mobile-phones-5507342/>

Computer Generations

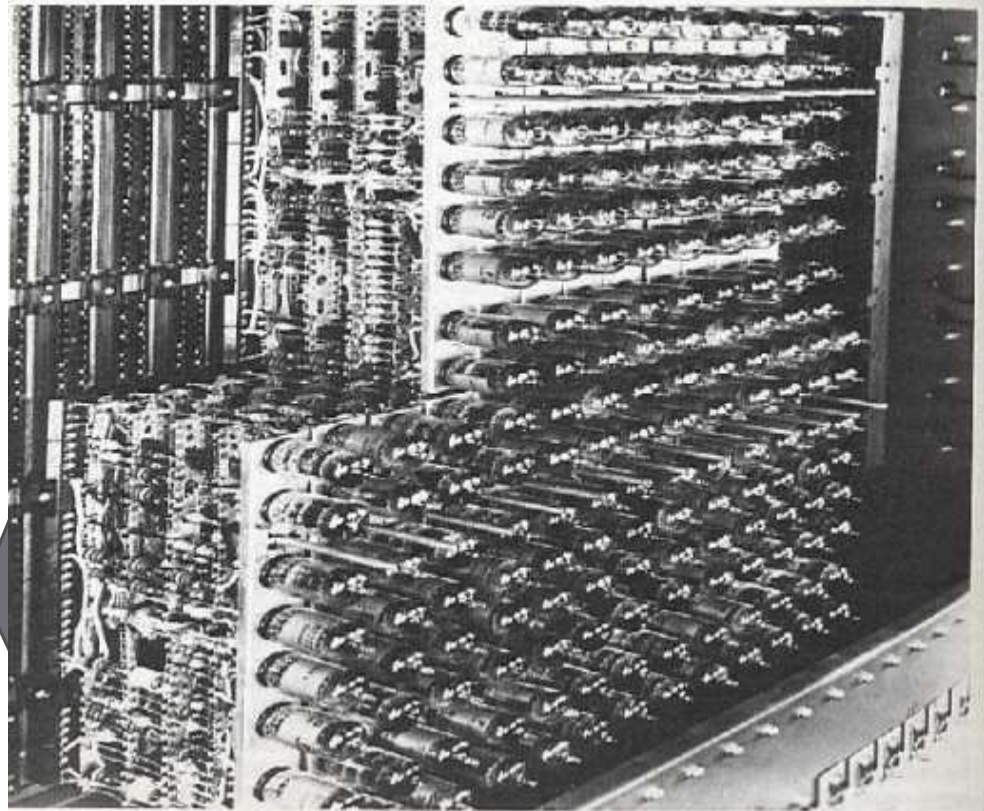


The First Generation: Vacuum Tubes



UNIVAC vacuum tube

<http://www.computermuseum.li/Testpage/UNIVAC-VacuumTube.htm>



IBM 701 panel

<http://www.columbia.edu/acis/history/701-tubes.html>

ENIAC

Background

- Electronic Numerical Integrator And Computer (ENIAC)
 - First **general-purpose** digital computer
 - Was built during WWII
- Eckert and Mauchly
 - University of Pennsylvania
- It was programmed to compute the values of **artillery range tables**
- It was the most powerful calculating device built to date
- It was the first programmable general-purpose electronic digital computer

ENIAC

Background

- Started 1943, finished 1946
 - Too late for war effort
 - Hence, used to study **hydrogen bomb** – regarded as general-purpose.
 - Its first task was doing calculations for the construction of a hydrogen bomb
- Used until 1955

ENIAC

Details

- Decimal (not binary)
- **20 accumulators of 10 digits**
- Programmed manually by switches
 - “Programming” is tedious
- **18,000 vacuum tubes**
- 30 tons
- 1500 square feet
- 140 kW power consumption
- **5,000 additions per second**



ENIAC

15,000 square feet



von Neumann/Turing

- “blueprint” to describe a machine
- A combination of von Neumann **architecture** and Turing **machine**

Von Neumann

- CPU – executing instructions and performing calculations
- Memory – stores instructions and data
- CU – Interprets and executes instructions
- ALU – Performs arithmetic and logical operations

Turing

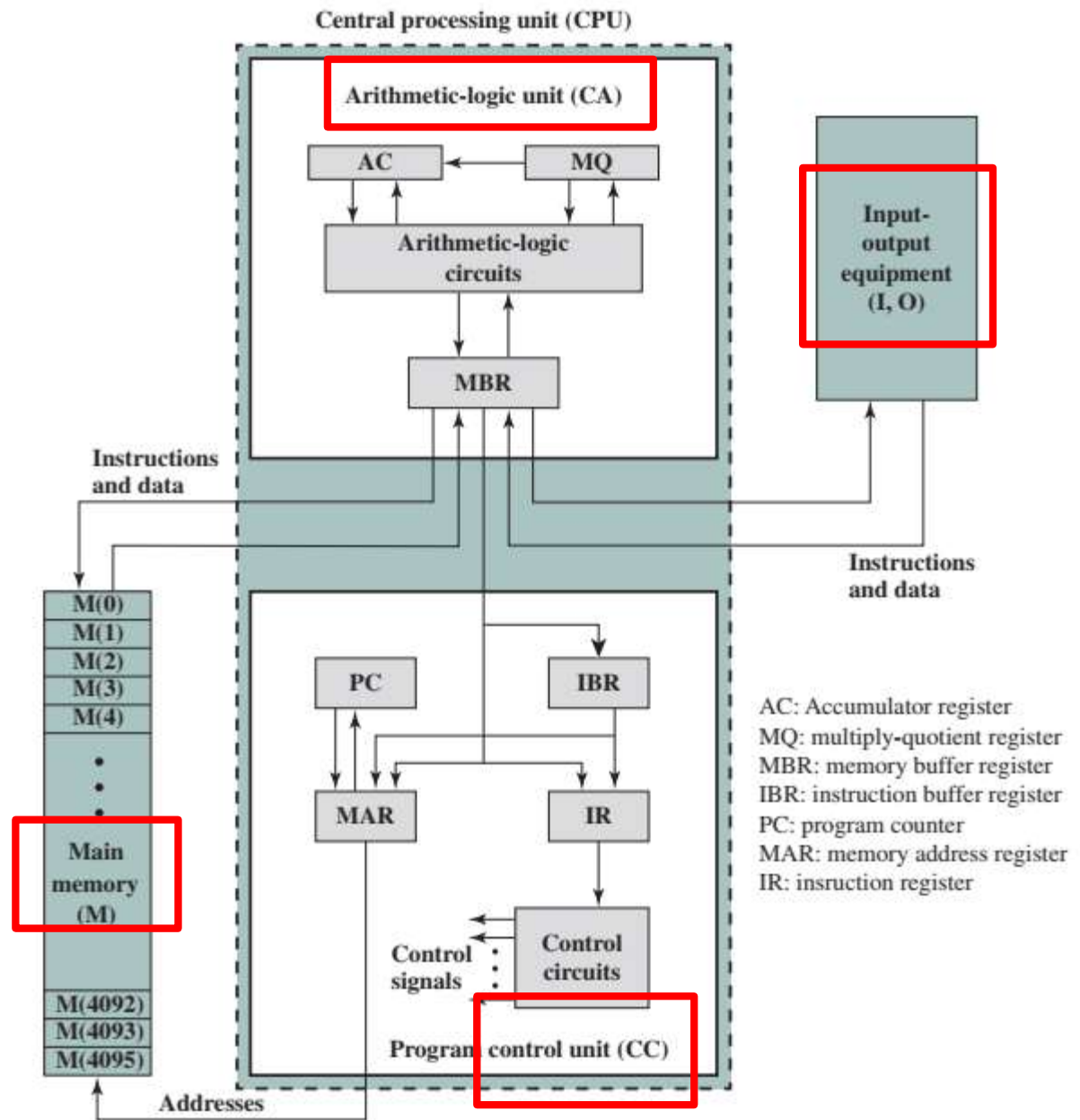
- Theoretical model of computation
- Simulate any algorithmic process and serves as a theoretical foundation

von Neumann/Turing

IAS

- was designed and built at the Institute for Advanced Study in the early 1950s by a team of researchers that included John von Neumann, Herman Goldstine, and J. Robert Oppenheimer.
- First stored-program computer concept (both instructions and data could be stored in the same memory)
- Designed at Princeton Institute for Advanced Studies
- The prototype of the **present computers**
- Started in 1946, completed in 1952
- IAS computer is known as the **stored-program concept**

IAS structure



IAS key concepts



Data and instructions both, are stored in a **single read-write memory**.



The contents of this memory are **addressable by location**.

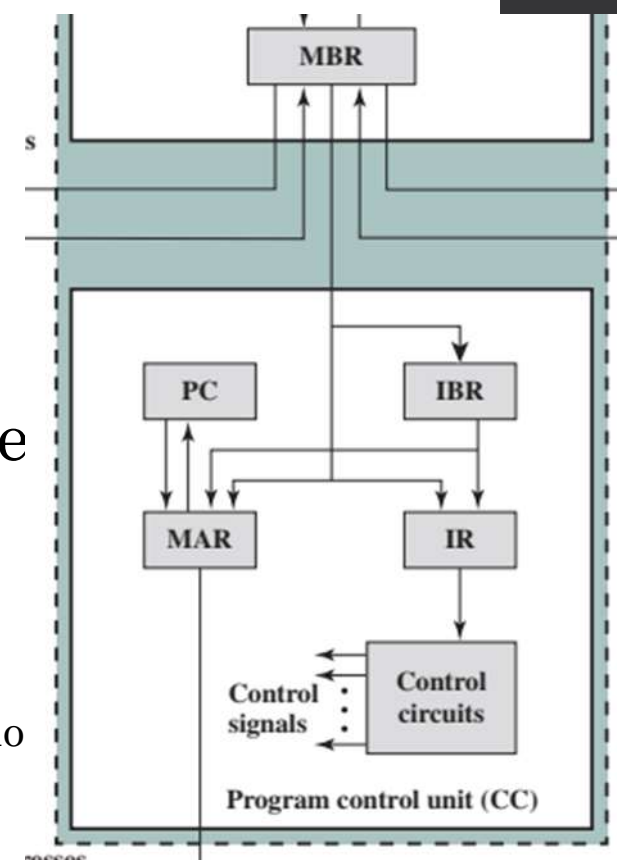


Execution occurs sequentially from one instruction to the next.

IAS Details

- IAS memory consists of 4096 storage
- **1000 x 40 bit words**
 - Binary number
 - 2 x 20-bit instructions
- CU operates the IAS by fetching instructions from memo them one at a time
- Set of **registers** (storage in CPU)
 - Memory Buffer Register (data)
 - Memory Address Register (address)
 - Instruction Register (instructions)
 - Instruction Buffer Register (temporary instructions)
 - Program Counter (address of next instruction)
 - Accumulator
 - Multiplier Quotient

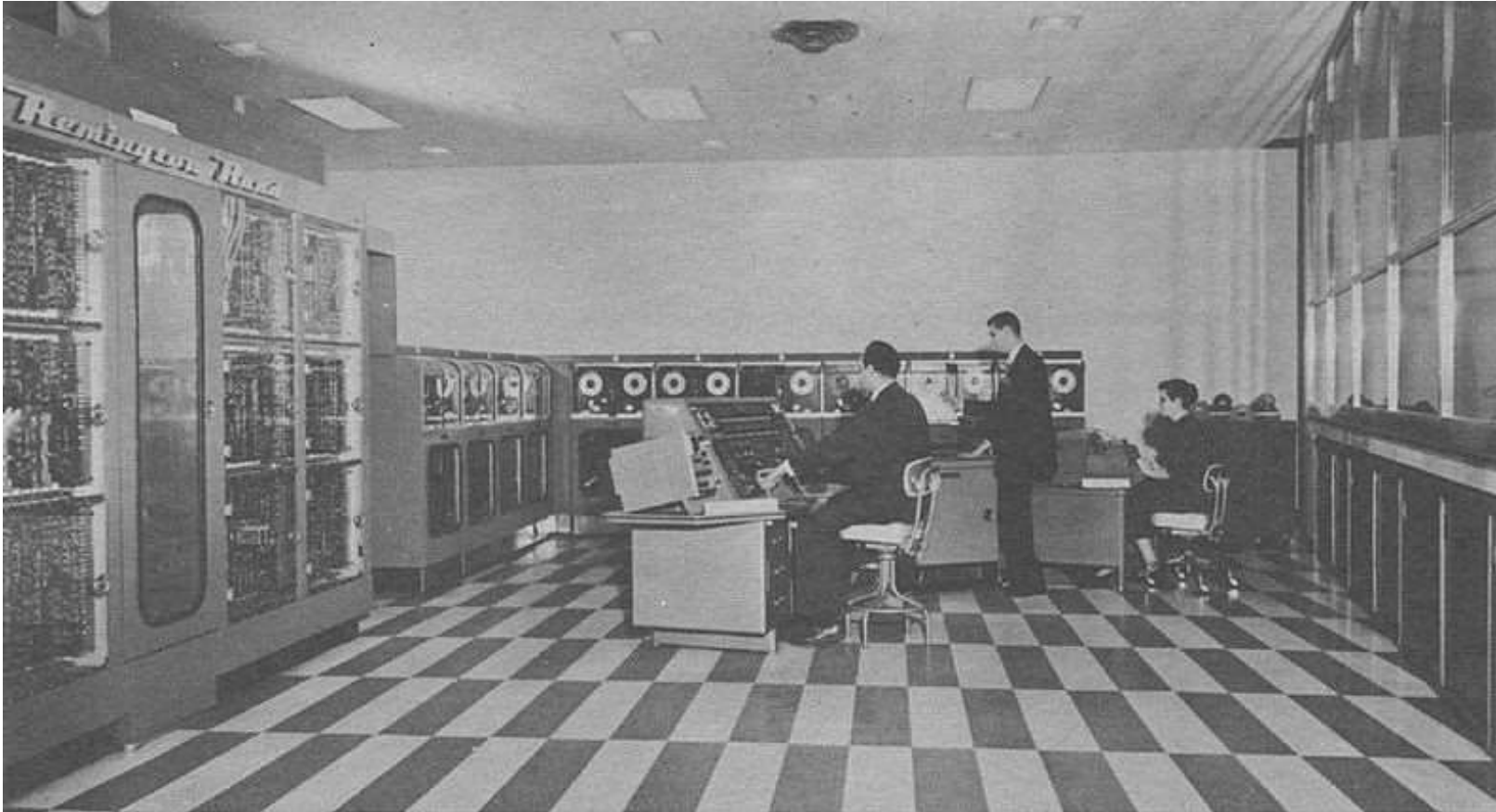
} Hold temporary operands and results of ALU operations



Commercial Computers

- 1947 - Eckert-Mauchly Computer Corporation
- UNIVAC I (Universal Automatic Computer)
 - First successful machine to be **commercialized**
 - Commissioned by US Bureau of Census 1950 calculations
- Became part of Sperry-Rand Corporation

UNIVAC I

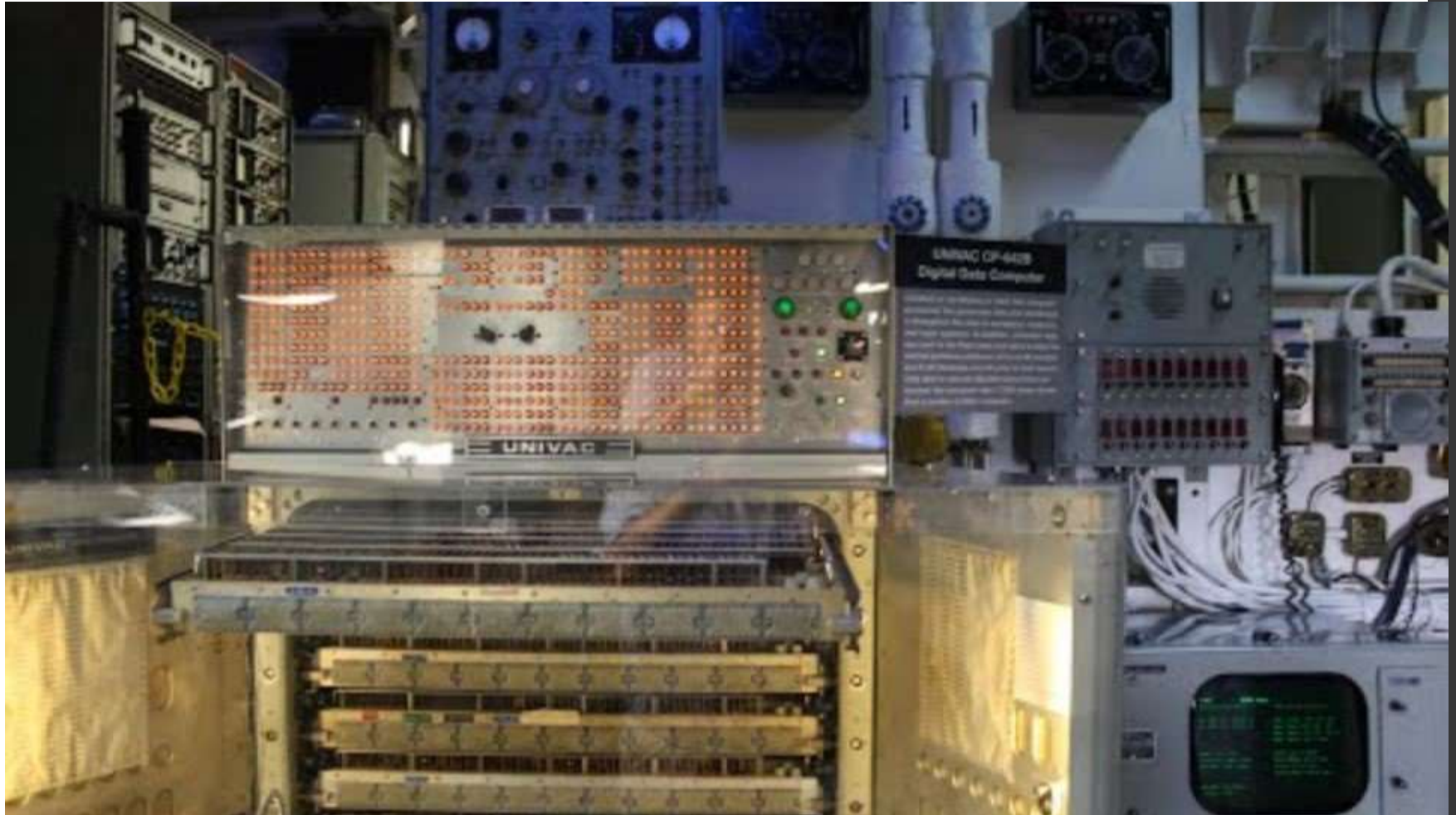


UNIVAC I at Franklin Life Insurance Company

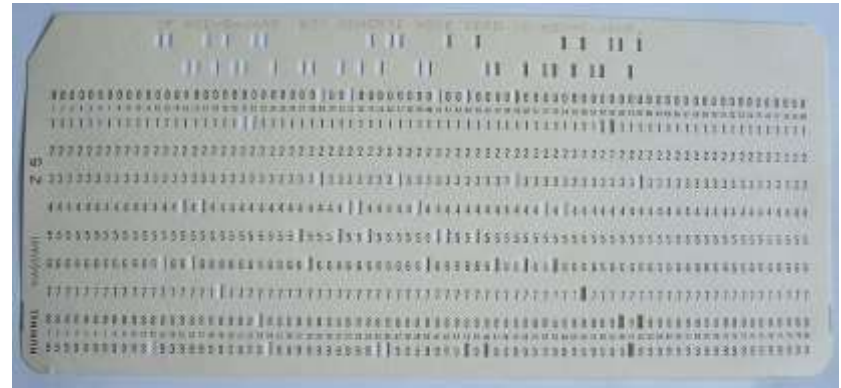
UNIVAC II

- Late 1950s
- **Faster** & more **memory**
 - Upward **compatibility** (programs on old machine can run on the new machines)
- UNIVAC 1100 series
 - Scientific applications

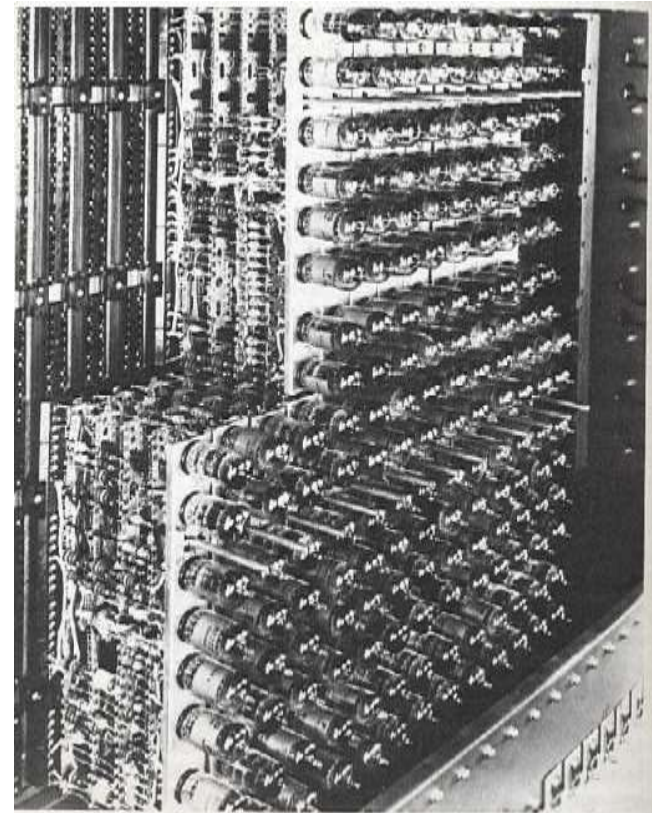
UNIVAC



IBM



- Major manufacturer of the **punched-card** processing equipment
- 1953 - the 701
 - IBM's first stored program computer
 - Scientific calculations
- 1955 - the 702
 - Business applications
- Lead to 700/7000 series



IBM 701 panel

Source: <https://commons.wikimedia.org/wiki/File:Punch-card-cobol.jpg>

<http://www.columbia.edu/acis/history/701-tubes.html>

IBM punched-card processing equipment



What is a vacuum tube?



The Second Generation: Transistors

- Replaced vacuum tubes
 - First major change for the electronic computer
- Transistors are:
 - Smaller
 - Cheaper
 - Less heat dissipation
 - Solid state device, made from silicon (sand)
- Invented 1947 at Bell Labs, William Shockley et al.



Transistors

<http://en.wikipedia.org/wiki/Transistor>

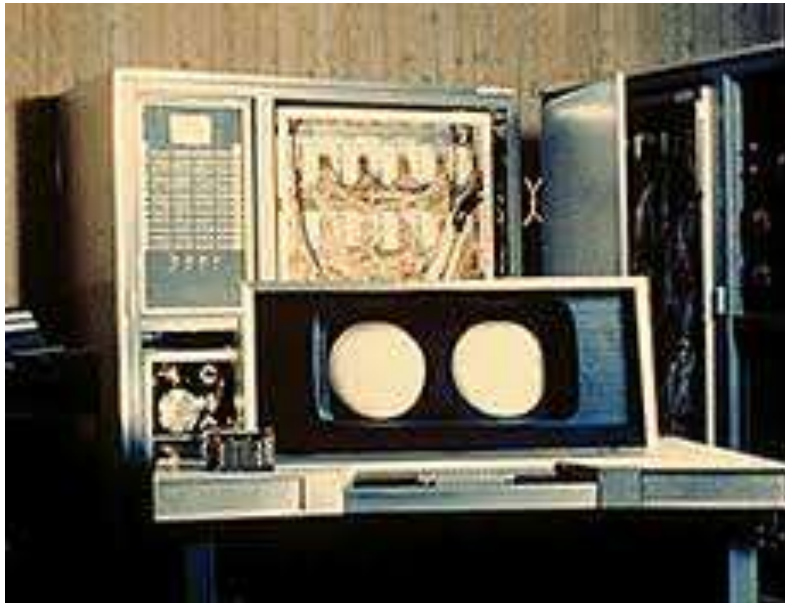
Transistor Based Computers

- Marked the **second-generation** machines
 - NCR & RCA produced small transistor machines
 - IBM 7000
 - DEC - 1957
 - Produced PDP-1
- Introduction of more **complex** ALU and control units

Transistor Based Computers

- Use **high-level** programming language
- Provided with **system software**
- Better performance, larger memory, smaller size
- But transistors (and other components) were discrete components (separate)
 - Problem when there are tens and hundreds of these components

Transistor Based Computers



IBM 7000



PDP-1

[PDP-1 Demo Video](#)

What is a transistor?



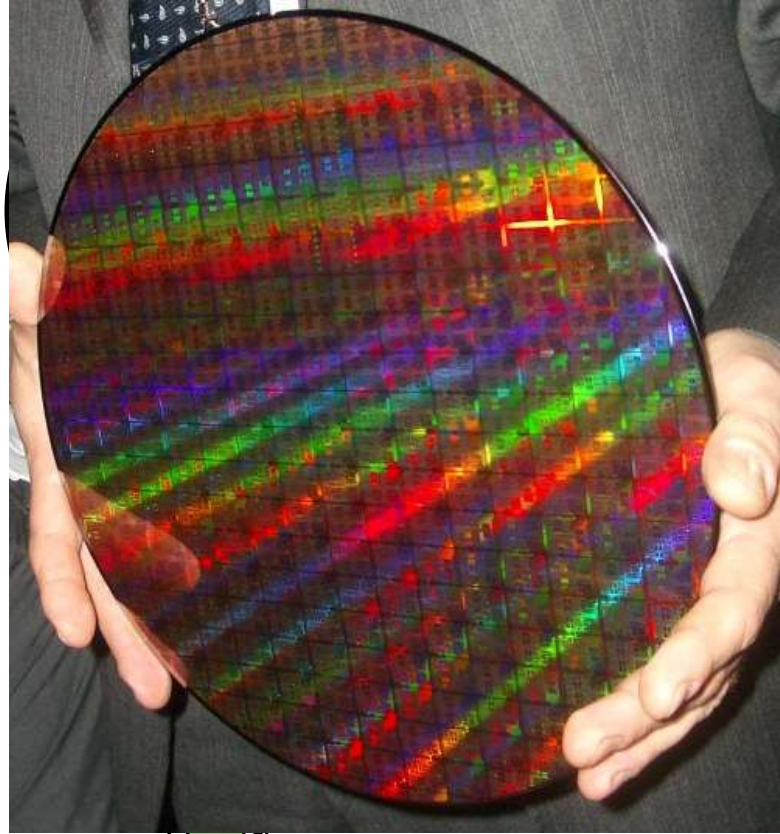
The Third Generation: Integrated Circuits

- Microelectronic
 - Literally means “small electronics”
- A computer is made up of:
 - **Gates**
 - Implement Boolean or logical function, control data flow
 - **Memory** cells
 - Stores 1 bit of data each
 - **Interconnections**

Integrated Circuits

- Remember the 4 functions?
 - Data **storage** (memory cells)
 - Data **processing** (gates)
 - Data **movements** (paths between components, through gates),
 - **Control** (paths carry control signal, activity is carried out with control signal)
- These can be manufactured on a semiconductor
 - E.g. silicon wafer

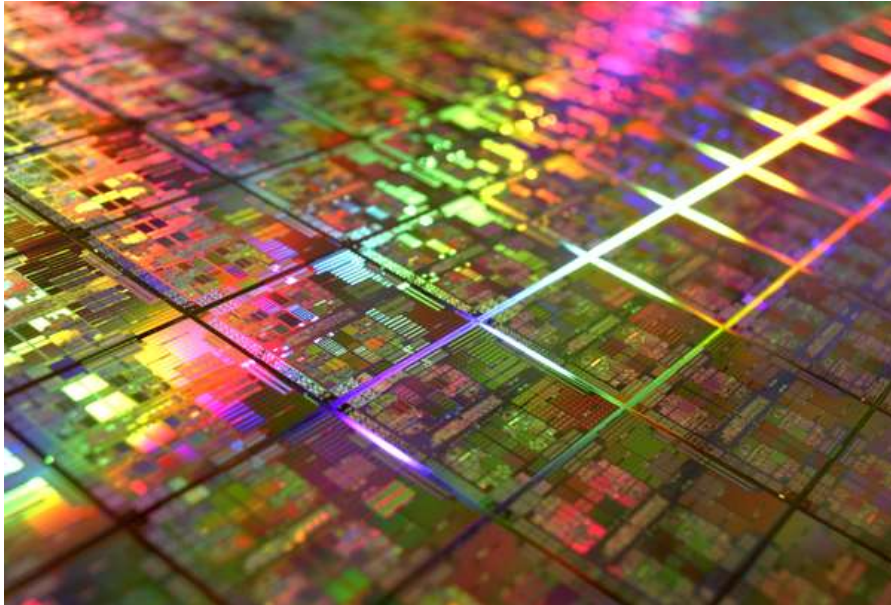
Integrated Circuits



AMD 45nm wafer

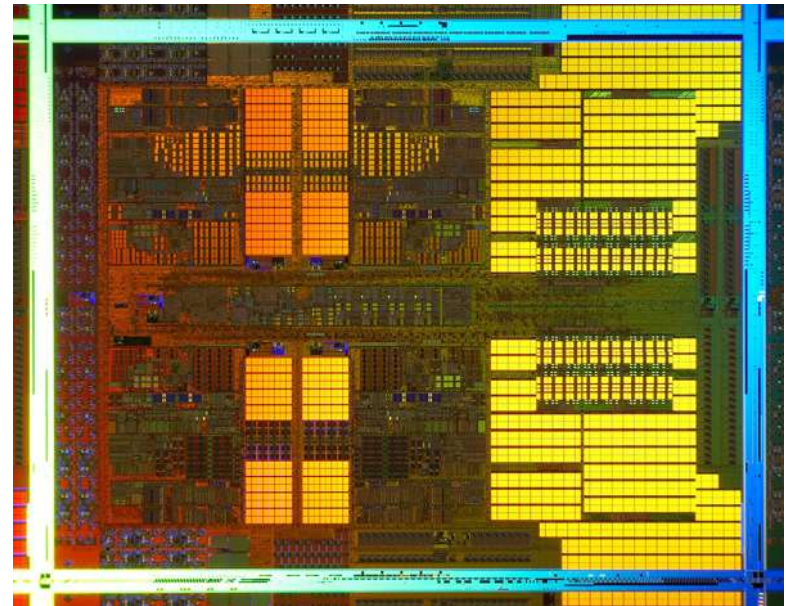
<http://www.theinquirer.net/inquirer/news/1005483/amd-shows-45nm>

Integrated Circuits



AMD 45nm Quad Core wafer

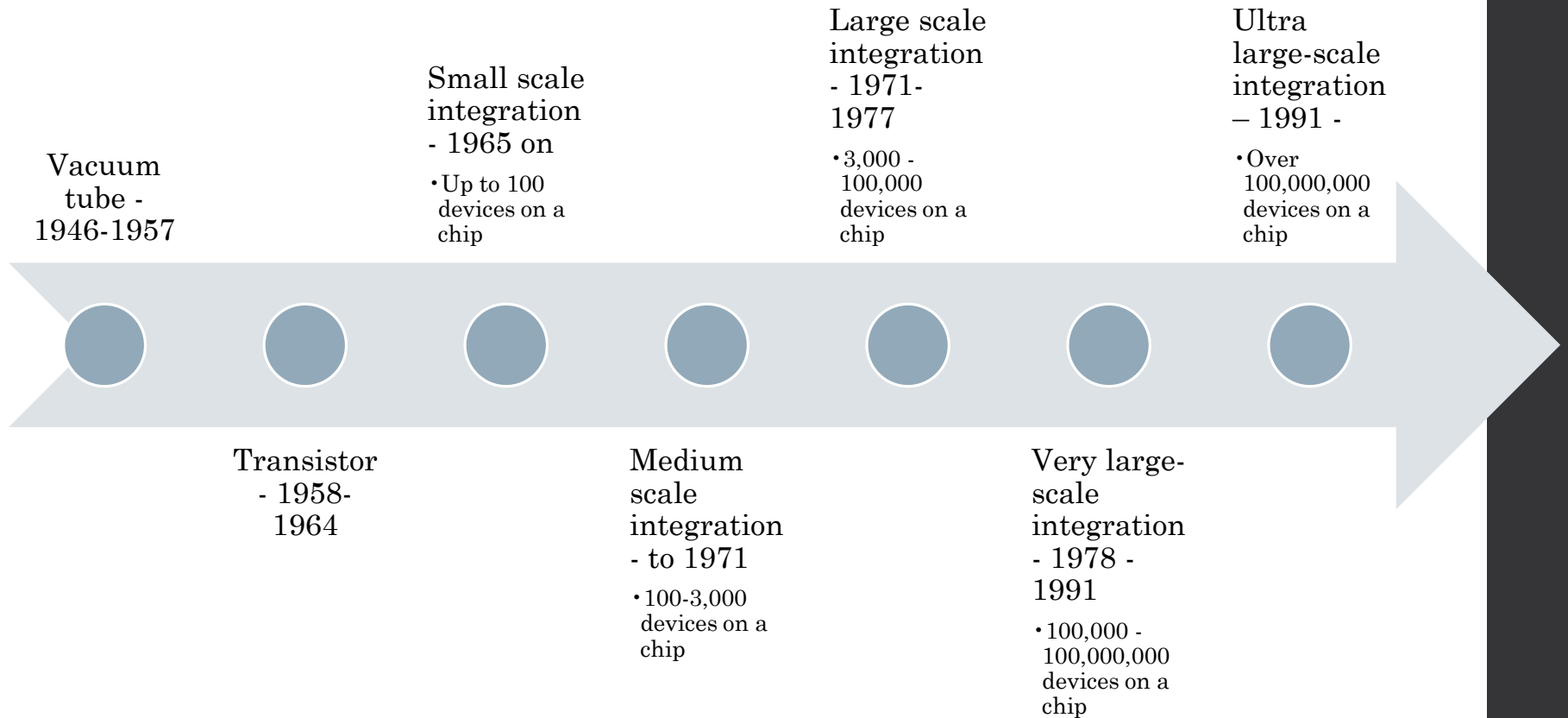
http://www.amd.com/us-en/0,,3715_15503,00.html



Single AMD 45nm Quad Core die

http://www.amd.com/us-en/0,,3715_15503,00.html

Generations of Computer



TIME 😊
FOR A
BREAK

Moore's Law

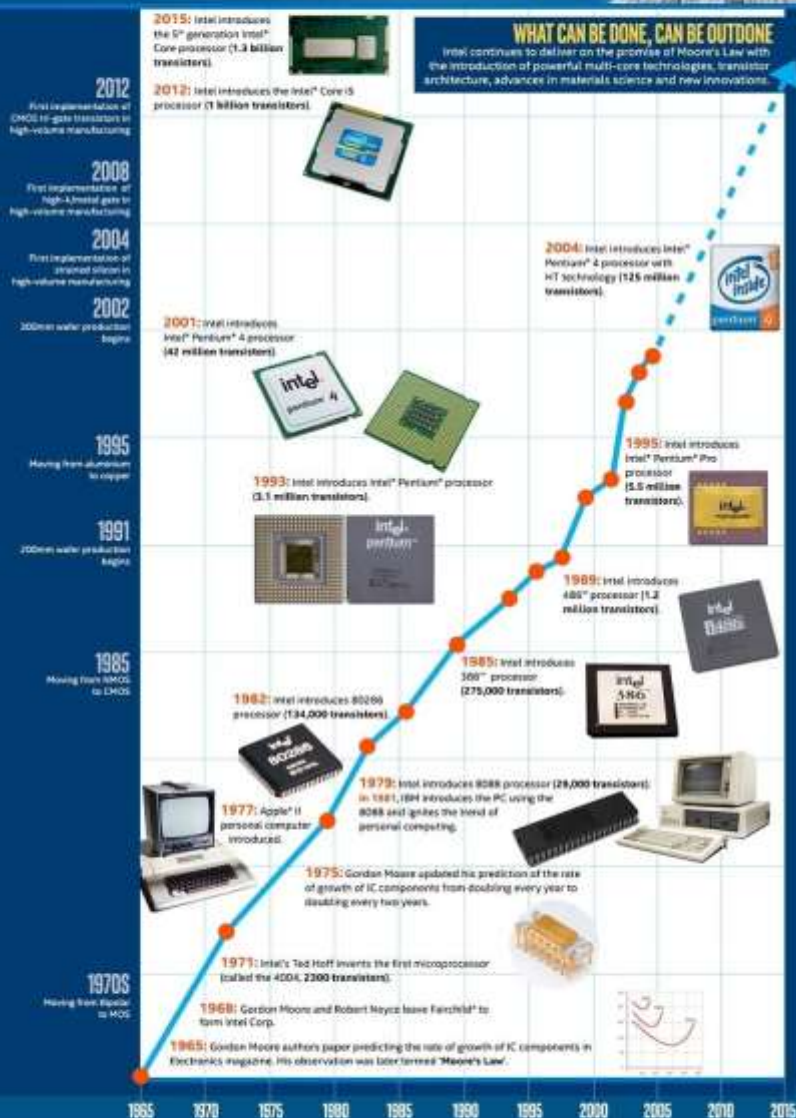
- Increased density of components on chip
- Gordon Moore – co-founder of Intel
- **Number of transistors** on a chip will **double every year**
- Since 1970's development has slowed a little
 - Number of transistors doubles **every 18 months**
- **Cost of a chip** has remained almost **unchanged**

Moore's Law

- Higher **packing density** means **shorter** electrical paths, giving higher **performance**
- Smaller size increased **flexibility**
- Reduced **power and cooling** requirements
- Fewer interconnections increase **reliability**

MOORE'S LAW TIMELINE

Moore's Law – the observation that computing dramatically decreases in cost at a regular pace – is short-hand for rapid technological change. Over the past 50 years, it has ushered in the dawn of the personalization of technology and enabled new experiences through the integration of technology into almost all aspects of our lives.

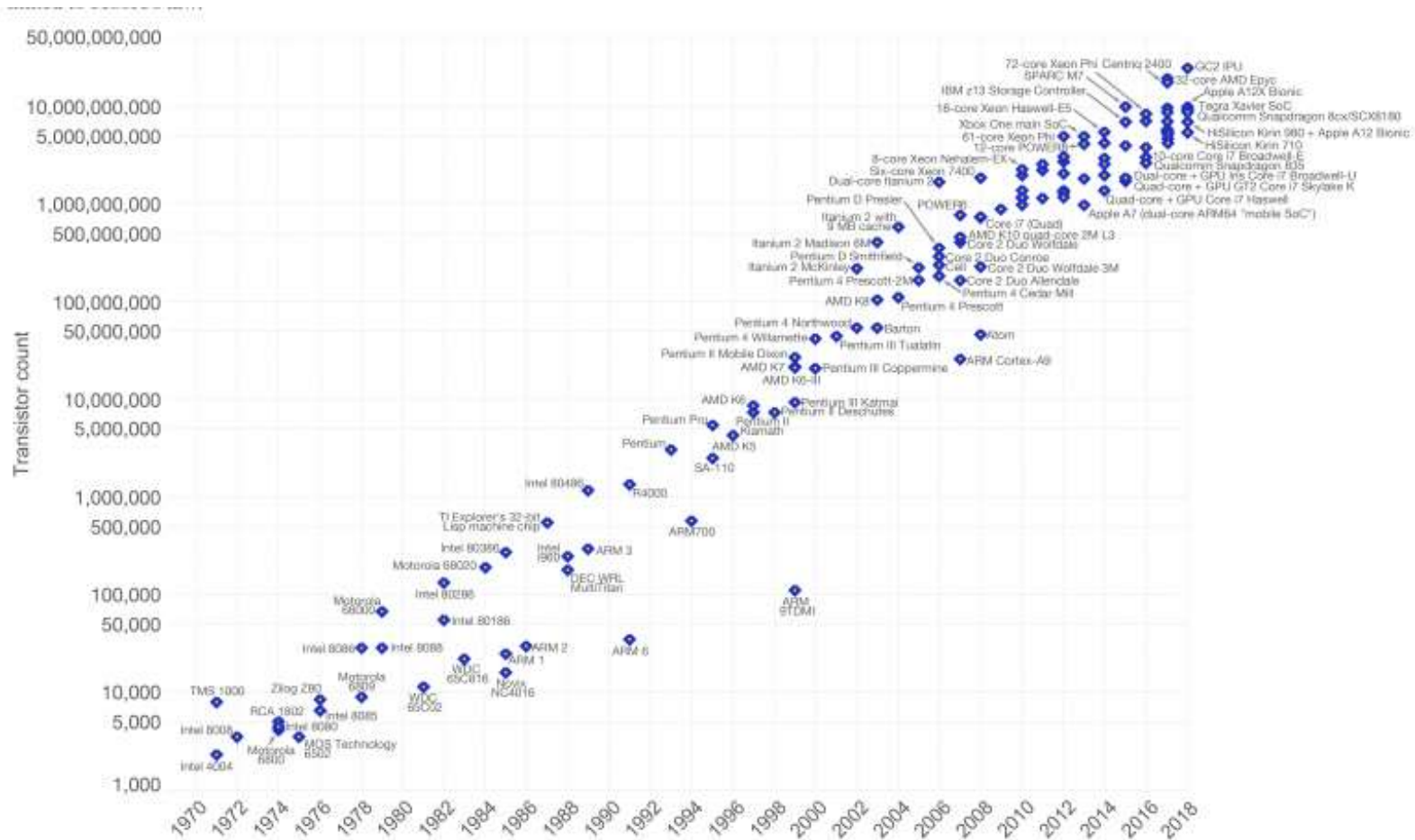


For more information, please visit intel.com.

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Growth in CPU Transistor Count



Data source: Wikipedia (https://en.wikipedia.org/wiki/Transistor_count)

The data visualization is available at [OurWorldInData.org](https://www.ourworldindata.org). There you find more visualizations and research on this topic.

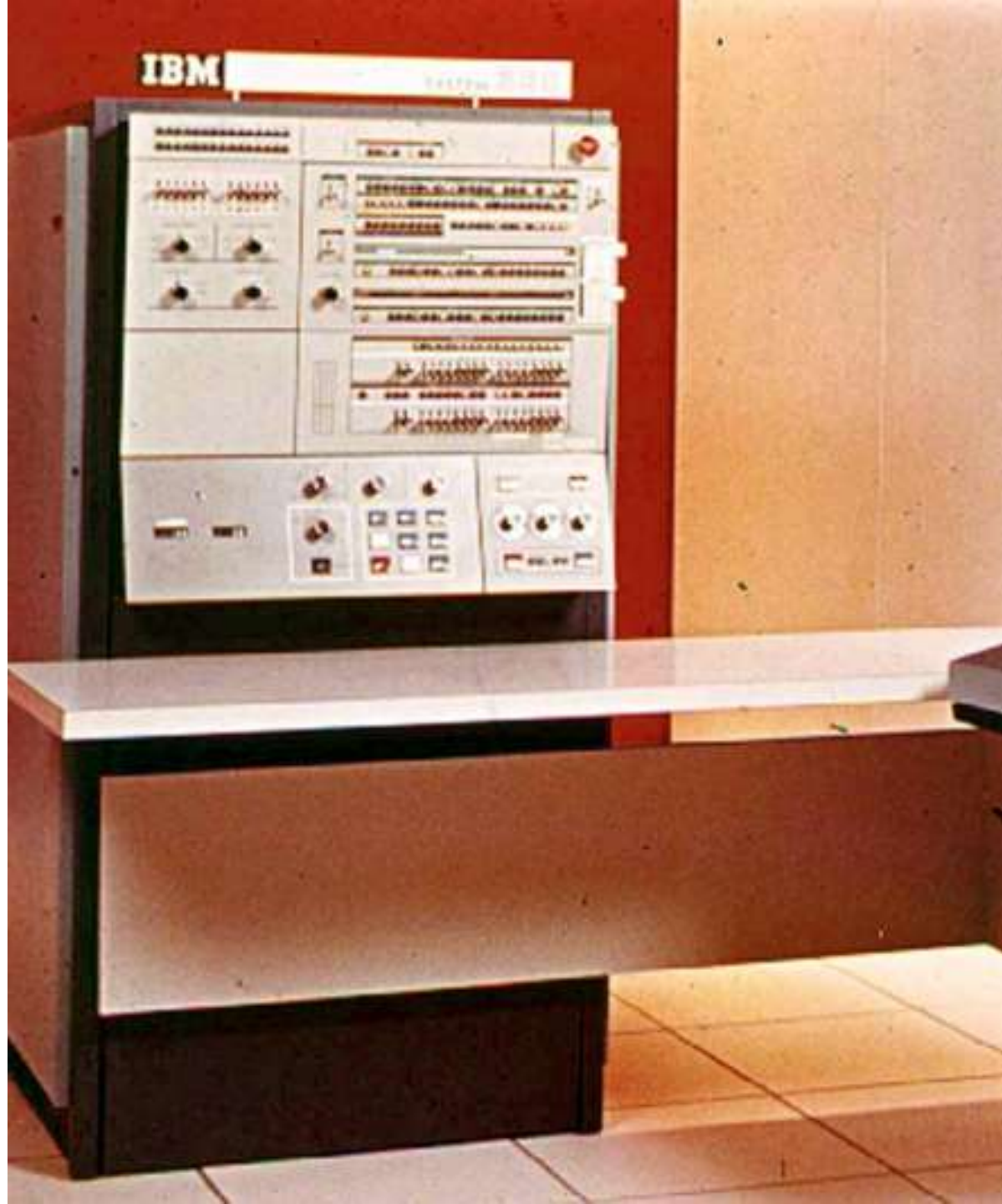
Licensed under CC-BY-SA by the author Max Roser.

Moore's Law



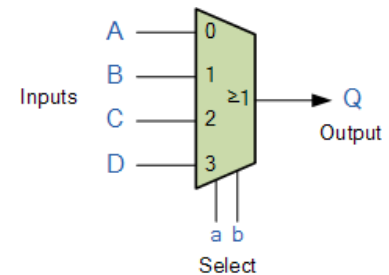
IBM 360 series

- A family of mainframe computer systems introduced by IBM in 1964
- Replaced (& not compatible with) 7000 series
- The **architecture** remains that of IBM's mainframe computers today



IBM 360 series

- First planned “family” of computers
 - Similar or identical instruction sets
 - Similar or identical O/S
 - Increasing speed
 - Increasing number of I/O ports (i.e. more terminals)
 - Increasing memory size
 - Increasing cost
- Multiplexed-switch structure (central-switched)

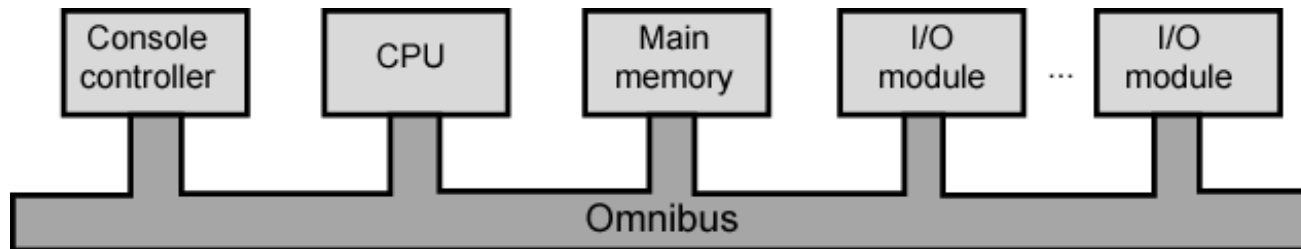


DEC PDP-8

- 1964
- First **minicomputer** (after miniskirt!)
- Did not need air-conditioned room
- **Small** enough to sit on a lab bench
- \$16,000
 - Compared to \$100k+ for IBM 360

DEC PDP-8

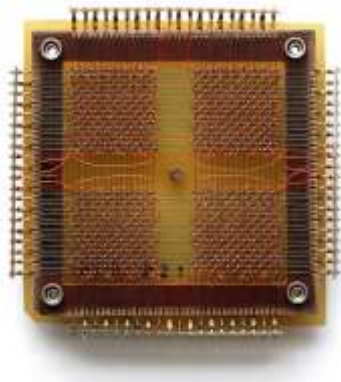
- Embedded applications & OEM
 - Another manufacturer can integrate PDP-8 into a total system for resale
- **Bus** structure





Later Generations: Semiconductor Memory

- ICs were initially used for processors (ALU, CU), **later used for memory**
- Previously memory made of **ferromagnetic material** (core) which is expensive, bulky, destructive readout



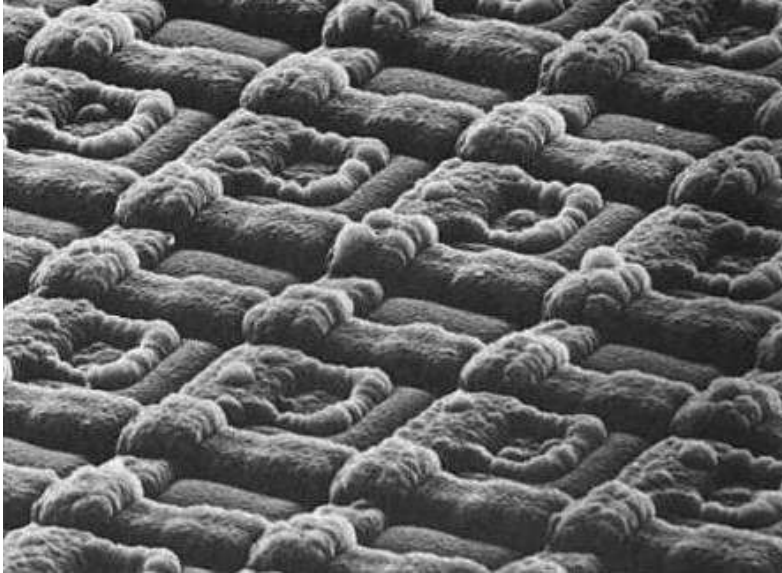
32 x 32 Core Memory

https://en.wikipedia.org/wiki/Magnetic-core_memory#/media/File:KL_CoreMemory.jpg

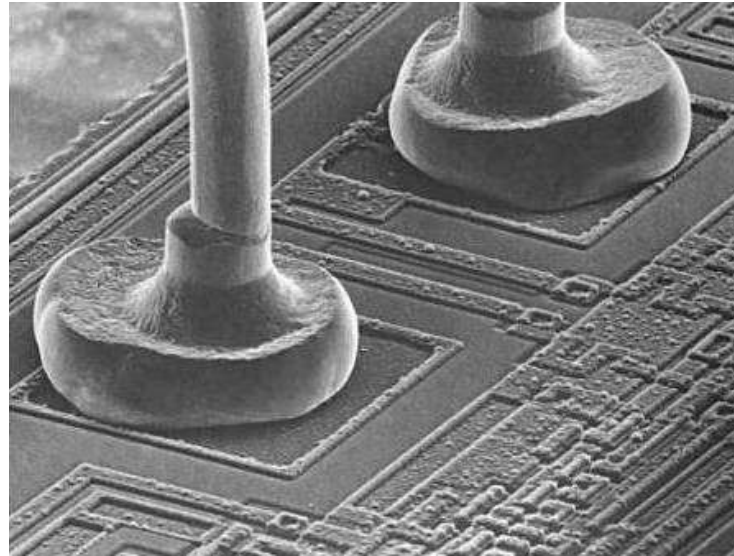
Semiconductor Memory

- Semiconductor Memory
 - 1970, Fairchild introduced a chip of the size of a single core (about 1.5mm) which could hold 256 bits
 - Non-destructive read & much faster than core but expensive
- Capacity approximately **doubles each year**

Later Generations: Semiconductor Memory



EPROM cells (transistors)

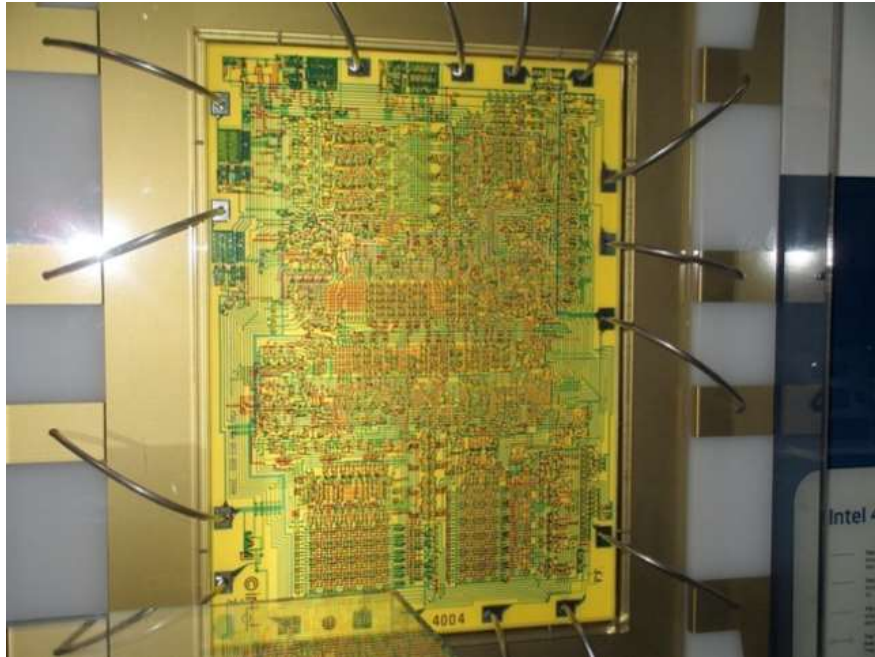
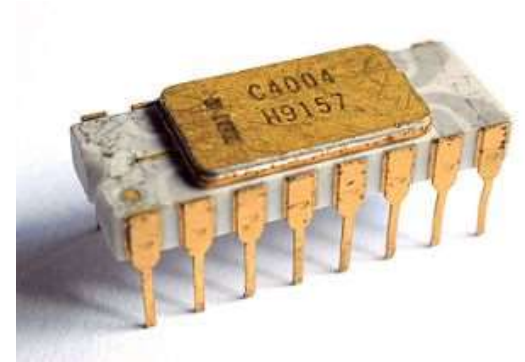


EPROM connecting wires

J. Burgess, M. Marten and R. Taylor, *Under the microscope: a hidden world revealed*, Cambridge University Press, 1990.

Microprocessors

- 1971 Intel introduced 4004
 - First microprocessor
 - All CPU components on a single chip
 - 4 bit



Source: https://commons.wikimedia.org/wiki/File:Intel_C4004.jpg
http://www.intel4004.com/current_intel_museum.htm

Microprocessors

- Followed in 1972 by 8008
 - 8 bit
 - Both designed for specific applications
- 1974 8080
 - Intel's first general purpose microprocessor

Microprocessors

- End 1970s powerful, general purpose 16-bit microprocessor appeared
 - 8086
- 1981 Bell Lab & Hewlett-Packard introduced 32-bit, single-chip processor
- 1985 80386

What computer did
you first get
acquainted with?

What did you use it
for?



Pair-Share



MOS 6502 chip

TIME 😊
FOR A
BREAK

Outline

- History of computers
- X86, PowerPC and ARM evolution

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and
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Structure and
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evolution and
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x86 Evolution

- 8080 (1974)
 - first general-purpose microprocessor
 - 8-bit data path
 - Used in first personal computer – Altair
- 8086 (1978)
 - much more powerful
 - 16 bit
 - instruction cache, prefetch few instructions
 - 8088 (8-bit external bus) used in first IBM PC
- 80286 (1982)
 - 16 Mbyte memory addressable
 - up from 1Mb



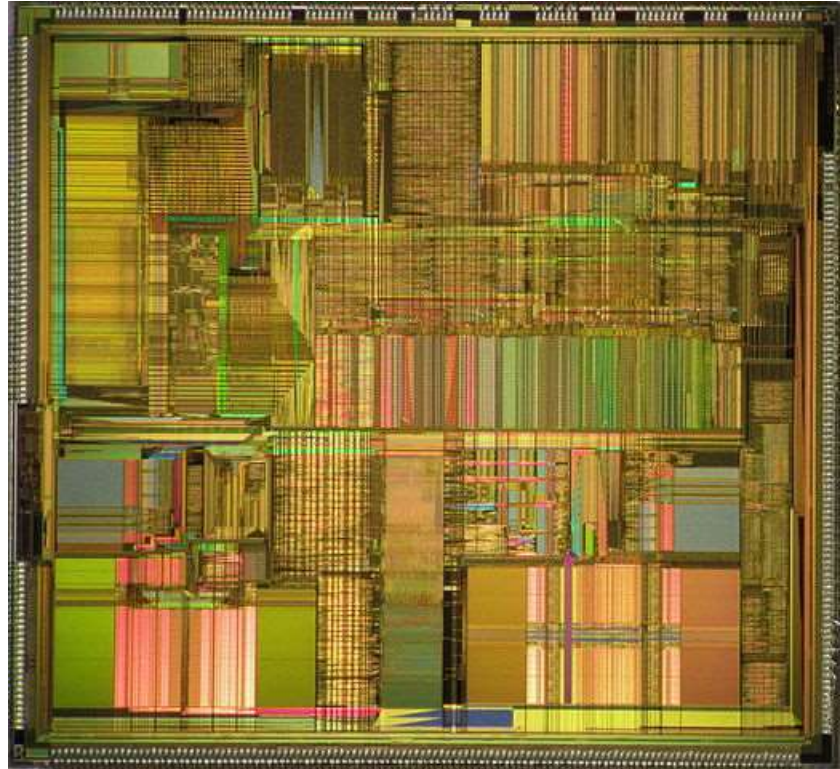
x86 Evolution

- 80386 (1985)
 - 32 bit
 - Support for **multitasking**
- 80486 (1989)
 - sophisticated powerful **cache** and **instruction pipelining**
 - built in **maths co-processor**



x86 Evolution

- Pentium (1993)
 - **Superscalar**
 - Multiple instructions executed in parallel



x86 Evolution

- Pentium Pro (1995)
 - Increased superscalar organization
 - Aggressive register renaming
 - branch prediction
 - data flow analysis
 - speculative execution
- Pentium II (1997)
 - MMX technology
 - graphics, video & audio processing
- Pentium III (1999)
 - Additional floating-point instructions for 3D graphics

x86 Evolution

- Pentium 4 (2000)
 - Note Arabic rather than Roman numerals
 - Further floating point and multimedia enhancements
- Core (2006)
 - First x86 with **dual core**

x86 Evolution

- Core 2 (2006)
 - **64-bit** architecture
- Core 2 Quad (2008)
 - 3GHz, 820 million transistors
 - **Four processors** on chip

x86 Evolution

	4004	8008	8080	8086	8088
Introduced	1971	1972	1974	1978	1979
Clock speeds	108 kHz	108 kHz	2 MHz	5 MHz, 8 MHz, 10 MHz	5 MHz, 8 MHz
Bus width	4 bits	8 bits	8 bits	16 bits	8 bits
Number of transistors	2,300	3,500	6,000	29,000	29,000
Feature size (µm)	10	8	6	3	6
Addressable memory	640 Bytes	16 KB	64 KB	1 MB	1 MB

(a) 1970s Processors

x86 Evolution

	80286	386TM DX	386TM SX	486TM DX CPU
Introduced	1982	1985	1988	1989
Clock speeds	6 MHz - 12.5 MHz	16 MHz - 33 MHz	16 MHz - 33 MHz	25 MHz - 50 MHz
Bus width	16 bits	32 bits	16 bits	32 bits
Number of transistors	134,000	275,000	275,000	1.2 million
Feature size (μm)	1.5	1	1	0.8 - 1
Addressable memory	16 MB	4 GB	16 MB	4 GB
Virtual memory	1 GB	64 TB	64 TB	64 TB
Cache	—	—	—	8 kB

(b) 1980s Processors

x86 Evolution

	486TM SX	Pentium	Pentium Pro	Pentium II
Introduced	1991	1993	1995	1997
Clock speeds	16 MHz - 33 MHz	60 MHz - 166 MHz,	150 MHz - 200 MHz	200 MHz - 300 MHz
Bus width	32 bits	32 bits	64 bits	64 bits
Number of transistors	1.185 million	3.1 million	5.5 million	7.5 million
Feature size (μm)	1	0.8	0.6	0.35
Addressable memory	4 GB	4 GB	64 GB	64 GB
Virtual memory	64 TB	64 TB	64 TB	64 TB
Cache	8 kB	8 kB	512 kB L1 and 1 MB L2	512 kB L2

(c) 1990s Processors

x86 Evolution

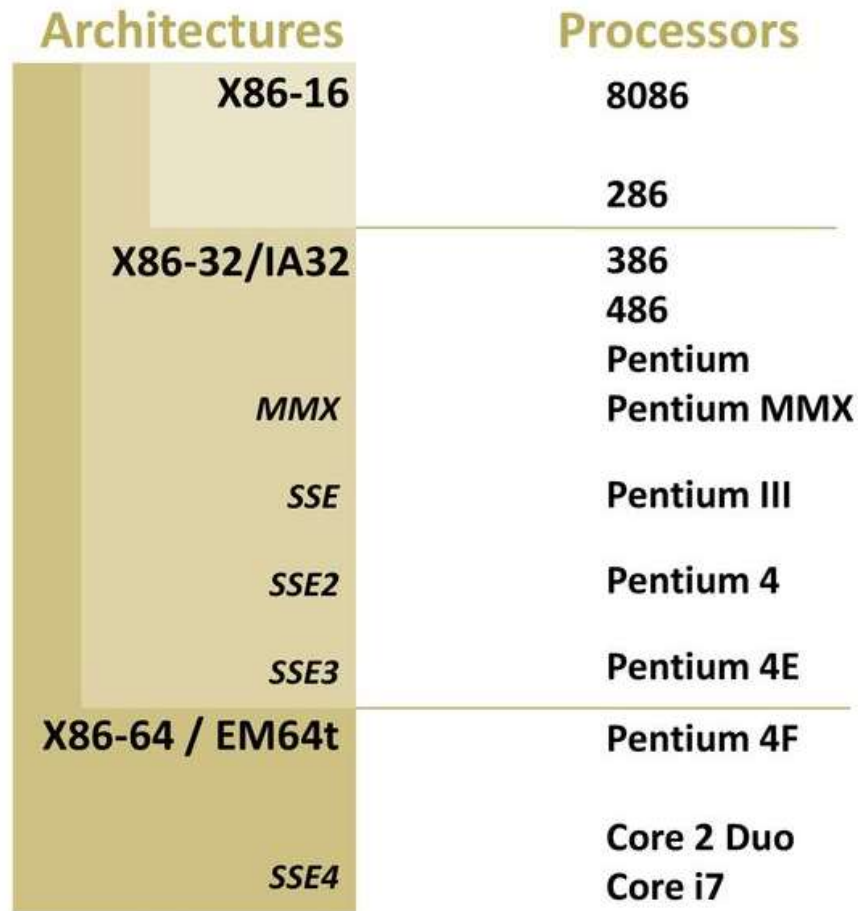
	Pentium III	Pentium 4	Core 2 Duo	Core i7 EE 4960X
Introduced	1999	2000	2006	2013
Clock speeds	450 - 660 MHz	1.3 - 1.8 GHz	1.06 - 1.2 GHz	4 GHz
Bus width	64 bits	64 bits	64 bits	64 bits
Number of transistors	9.5 million	42 million	167 million	1.86 billion
Feature size (nm)	250	180	65	22
Addressable memory	64 GB	64 GB	64 GB	64 GB
Virtual memory	64 TB	64 TB	64 TB	64 TB
Cache	512 kB L2	256 kB L2	2 MB L2	1.5 MB L2/15 MB L3
Number of cores	1	1	2	6

(d) Recent Processors

x86 Evolution

- x86 architecture dominant **outside embedded** systems
- Organization and technology changed dramatically
- Instruction set architecture evolved with backwards compatibility
- ~1 instruction per month added
- 500 instructions available
- See Intel web pages for detailed information on processors

Intel x86 Processors: Overview



PowerPC

- 1975, 801 minicomputer project (IBM) **RISC**
- Berkeley RISC I processor
- 1986, IBM commercial RISC workstation product, RT PC.
 - Not commercial success
 - Many rivals with comparable or better performance
- 1990, IBM RISC System/6000
 - RISC-like superscalar machine
 - POWER architecture

PowerPC

- IBM alliance with Motorola (68000 microprocessors), and Apple, (used 68000 in Macintosh)
- Result is **PowerPC architecture**
 - Derived from the POWER architecture
 - Superscalar RISC
 - Apple Macintosh
 - Embedded chip applications

PowerPC Family

- 601:
 - Quickly to market. 32-bit machine
- 603:
 - Low-end desktop and portable
 - 32-bit
 - Comparable performance with 601
 - Lower cost and more efficient implementation
- 604:
 - Desktop and low-end servers
 - 32-bit machine
 - Much more advanced superscalar design
 - Greater performance

PowerPC Family

- 620:
 - High-end servers
 - 64-bit architecture
- 740/750:
 - Also known as G3
 - Two levels of cache on chip
- G4 (1999-2004):
 - Increases parallelism and internal speed
- G5 (2003-2006):
 - Improvements in parallelism and internal speed
 - **64-bit** organization

- Although the organization and technology of the x86 machines have changed dramatically over the decades, the instruction set architecture has evolved to remain backward compatible with earlier versions.
- Any program written on an older version of the x86 architecture can execute on newer versions



Embedded Systems

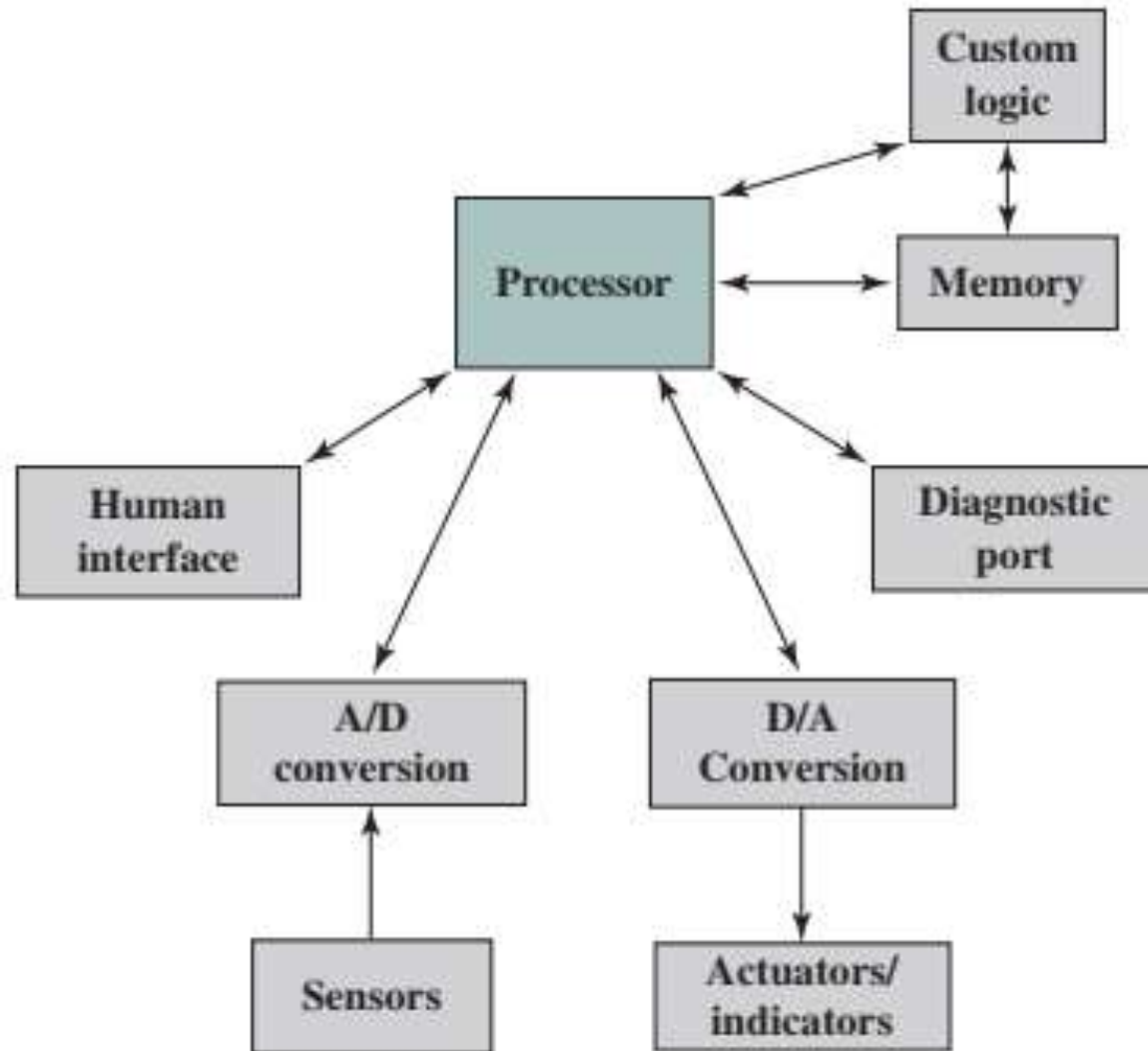
ARM

- ARM evolved from **RISC** design
- Used mainly in **embedded** systems
 - Used **within** product
 - Not general-purpose computer
 - Dedicated function
 - E.g. Anti-lock brakes in car, household appliances, automotive airbag

Embedded Systems Requirements

- Different **sizes**
 - Different constraints, optimization, reuse
- Different **requirements**
 - Safety, reliability, real-time, flexibility, legislation
 - Lifespan
 - Environmental conditions
 - Static v dynamic loads
 - Slow to fast speeds
 - Computation v I/O intensive
 - Discrete event v continuous dynamics

Possible Organization of an Embedded System

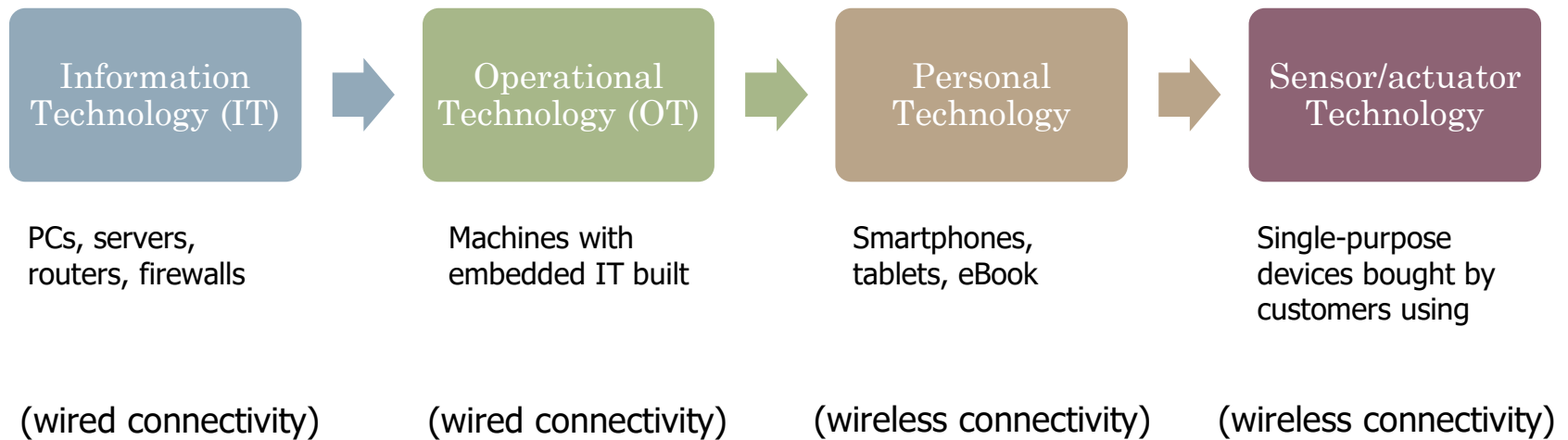


The Internet of Things (IoT)

- Expanding interconnection of **smart devices**, ranging from **appliances** to tiny **sensors**
- Driven by deeply embedded devices.



IoT evolution



ARM Evolution

- Designed by ARM Inc., Cambridge, England
- Licensed to manufacturers
- **High speed, small die size, low power consumption**

ARM Evolution

- PDAs, handheld games, phones
 - E.g. iPod, iPhone
- Acorn produced ARM1 & ARM2 in 1985 and ARM3 in 1989
- Acorn, VLSI and Apple Computer founded ARM Ltd.

ARM Products

Cortex-A

- Cortex-A5
- Cortex-A7
- :
- :
- Cortex-A77
- Cortex-A78

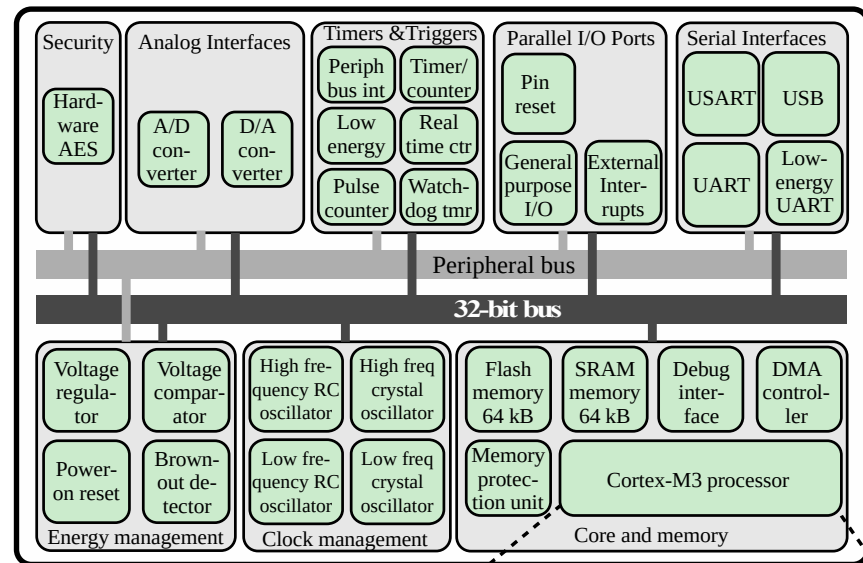
Cortex-R

- Cortex-R4(F)
- Cortex-R5(F)
- :
- :
- Cortex-R52(F)
- Cortex-R82(F)

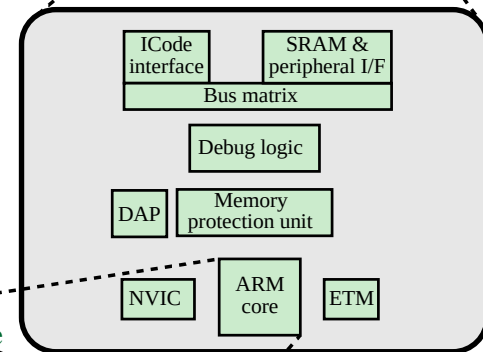
Cortex-M

- Cortex-M0
- Cortex-M0+
- Cortex-M3
- Cortex-M4
- :
- :
- Cortex-M55

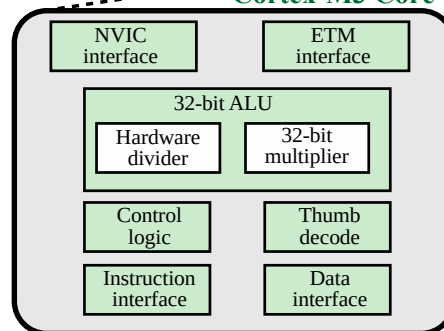
ARM Cortex-M3-based Microcontroller



Microcontroller Chip



Cortex-M3 Processor



Cortex-M3 Core

Quick Research

Cortex-A

Cortex-R

Cortex-M

- What's the difference between the Cortex A, R, and M series?



ARM Systems Categories

- Embedded **real time**, safety-critical
- **Application** platform
 - Linux, Palm OS, Symbian OS, Windows mobile, Android
- **Microcontrollers**

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Cloud Computing

- Internet-connected **infrastructure**
- Model for enabling **ubiquitous**,
convenient, **on-demand** network
access to a **shared pool** of configurable
computing resources... (NIST)

Cloud Services

- Software as a Service (SaaS)
 - **Application software** running on and accessible in the cloud.
 - E.g. Google Apps, Microsoft Office 365, Dropbox, ...

Cloud Services

- Platform as a Service (PaaS)
 - Platform on which customer's applications can run.
 - Software **building blocks**, development/programming **tools**, and **environments to deploy** new applications.
 - E.g. Windows Azure, Google App Engine, ...

Cloud Services

- Infrastructure as a Service (IaaS)
 - Access to cloud infrastructure.
- **Processing**, **storage**, **networks**, and other resources.
- E.g. Windows Azure, Google Compute Engine, Amazon Web Services, ...

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Performance Assessment

Key parameters

Performance

Cost

Size

Security

Reliability

Power
consumption

Clock Speed

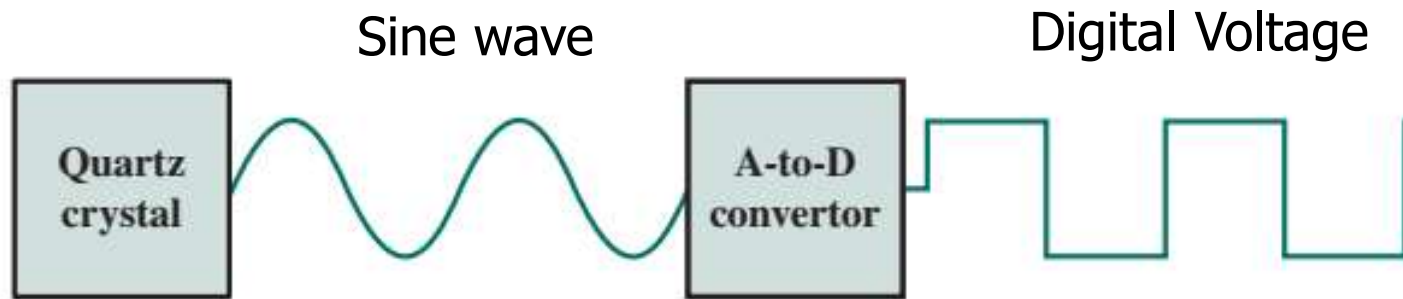
- The speed of a processor is dictated by the pulse frequency produced by the clock, measured in
 - cycles per second
 - Hertz (Hz)
- Clock signals are generated by a quartz crystal, which generates a constant sine wave while power is applied. This wave is converted into a digital voltage
- Signals in CPU take time to settle down to 1 or 0
- Signals may change at different speeds
- Operations need to be synchronised
- Instruction execution in discrete steps
 - Fetch, decode, load and store, arithmetic or logical
 - Usually require multiple clock cycles per instruction
- Pipelining gives simultaneous execution of instructions

Clock Speed

Rate of pulse =
clock rate
/clock speed

One increment
of pulse =
clock cycle

Time between
pulses = cycle
time

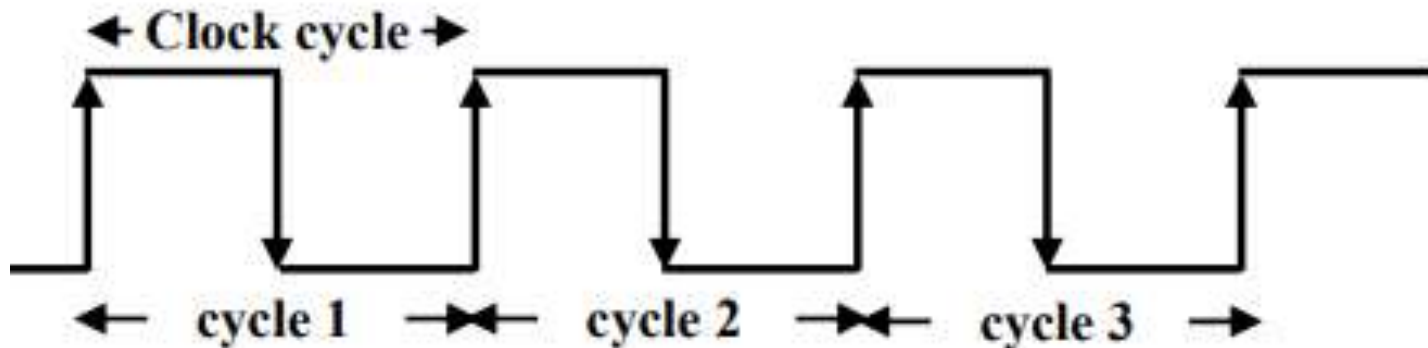


System Clock

Instruction Execution Rate

- Most computers run synchronously utilizing a CPU clock running at a constant clock rate, where clock rate = $1/f$

$$t = 1/f$$



- The CPU clock rate depends on the specific CPU organization (design) and hardware implementation technology (VLSI) used
- A single machine instruction may take one or more CPU cycles to complete termed as the **Cycles Per Instruction (CPI)**.

Cycle per instruction (CPI)

- Cycle/sec = Hertz = Hz
- MHz = 10^6 Hz
- GHz = 10^9 Hz

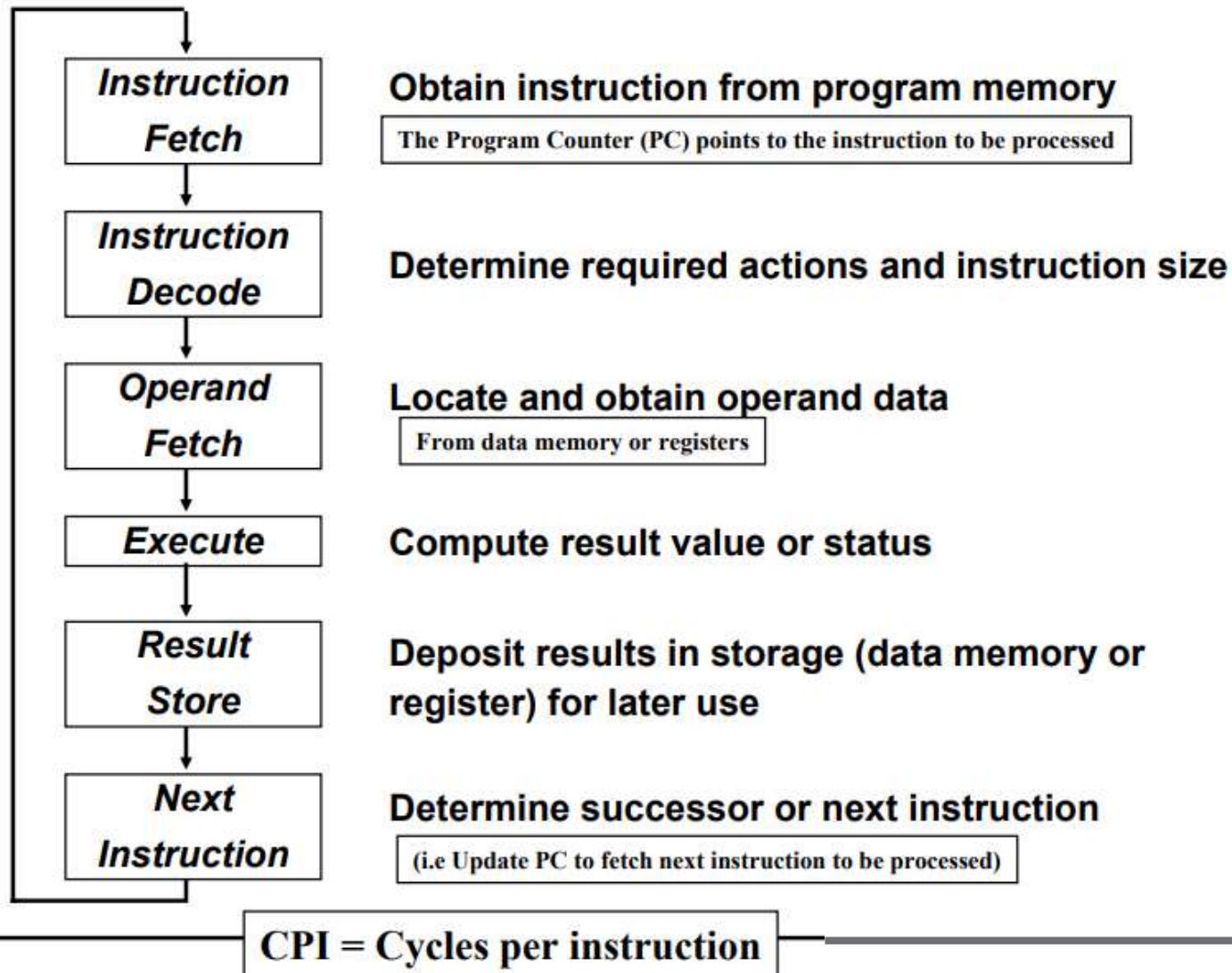
Average instruction (CPI)

- What's an “average instruction” (CPI)?
- Given a program, how many cycles does an instruction typically take?

Depends on how many instructions and what are the types of instructions

E.g. all addition vs all loads for multi-cycle implementation

Generic CPU Machine Instruction Processing Steps



Instruction Execution Rate

- A **common measure of performance for a processor** is the rate at which instructions are executed, expressed as millions of instructions per second (MIPS)

$$\text{MIPS rate} = \frac{I_c}{T \times 10^6} = \frac{f}{CPI \times 10^6}$$

MIPS calculation

% total instructions
count corresponding
to each instruction
class

- Consider the execution of a program that results in the execution of 2 million instructions on a 400-MHz processor. The program consists of four major types of instructions. The instruction mix and the CPI for each instruction type are given below, based on the result of a program trace experiment:

Instruction Type	<i>CPI</i>	Instruction Mix (%)
Arithmetic and logic	1	60
Load/store with cache hit	2	18
Branch	4	12
Memory reference with cache miss	8	10

Step 1: Calculate the CPI

$$\text{CPI} = (1 \times 0.6) + (2 \times 0.18) + (4 \times 0.12) + (8 \times 0.1) = 2.24$$

Step 2: Calculate the MIPS

$$\text{MIPS rate} = (400 \times 10^6) / (2.24 \times 10^6) \sim 178$$

Quick activity

- Calculate the MIPS based on the information given below:

Instruction type	CPI	Instruction Mix (%)
Fetch	2	12
Load/store	4	18
Decode	2	20
Arithmetic & logic	5	50

Benchmarks

- Measures such as MIPS and MFLOPS have proven inadequate to evaluating the performance of processors.
- Because of differences in instruction sets, the instruction execution rate is not a valid means of comparing the performance of different architectures
- Alternatively, we use benchmarks: **Programs** designed to test performance

Written in high
level language

Represents style
of task - Systems,
numerical,
commercial

Easily measured

Widely
distributed

Benchmarks: SPEC

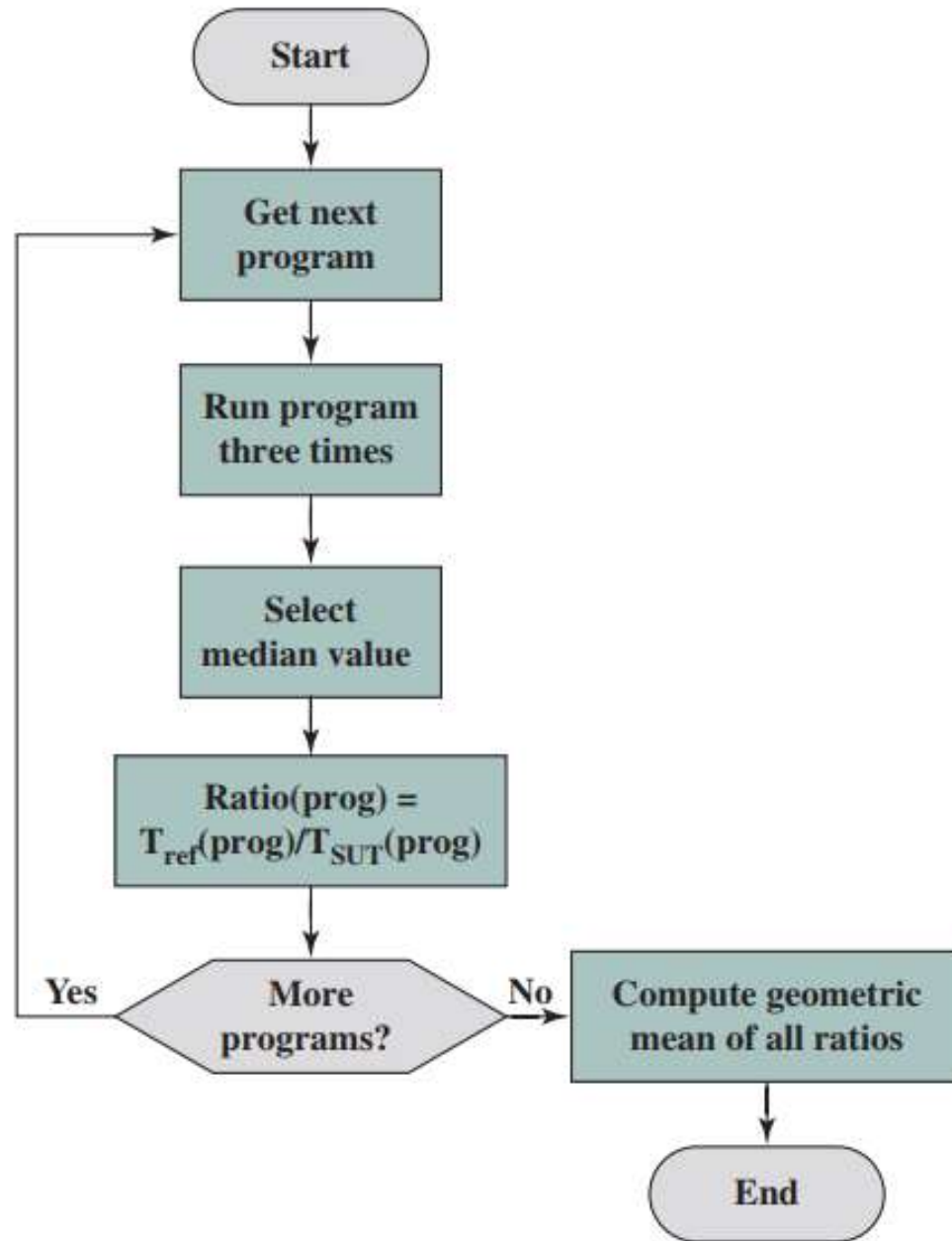
- System Performance Evaluation Corporation (SPEC)
 - CPU2017 for computation bound
 - floating point programs
 - integer programs
 - Speed and rate metrics
 - Single task and throughput
- The best known of the SPEC benchmark suites is SPEC CPU2017.
- This is the industry standard suite for processor-intensive applications.

<https://www.spec.org/benchmarks.html>

SPEC suites

- **SPEC Cloud_IaaS** - addresses the performance of infrastructure-as-a-service (IaaS) public or private cloud platforms
- **SPECviewperf** - measuring 3D graphics performance based on professional applications
- **SPECwpc** - measure all key aspects of workstation performance based on diverse professional applications, including media and entertainment, product development, life sciences, financial services, and energy
- **SPECjvm2008** - evaluate performance of the combined hardware and software aspects of the Java Virtual Machine (JVM) client platform
- **SPECjbb2015** (Java Business Benchmark) - evaluating server-side Java-based electronic commerce applications.
- **SPECsfs2014** - evaluate the speed and request-handling capabilities of file servers
- **SPECvirt_sc2013** - Performance evaluation of datacenter servers used in virtualized server consolidation. Measures the end-to-end performance of all system components

SPEC Evaluation Flowchart

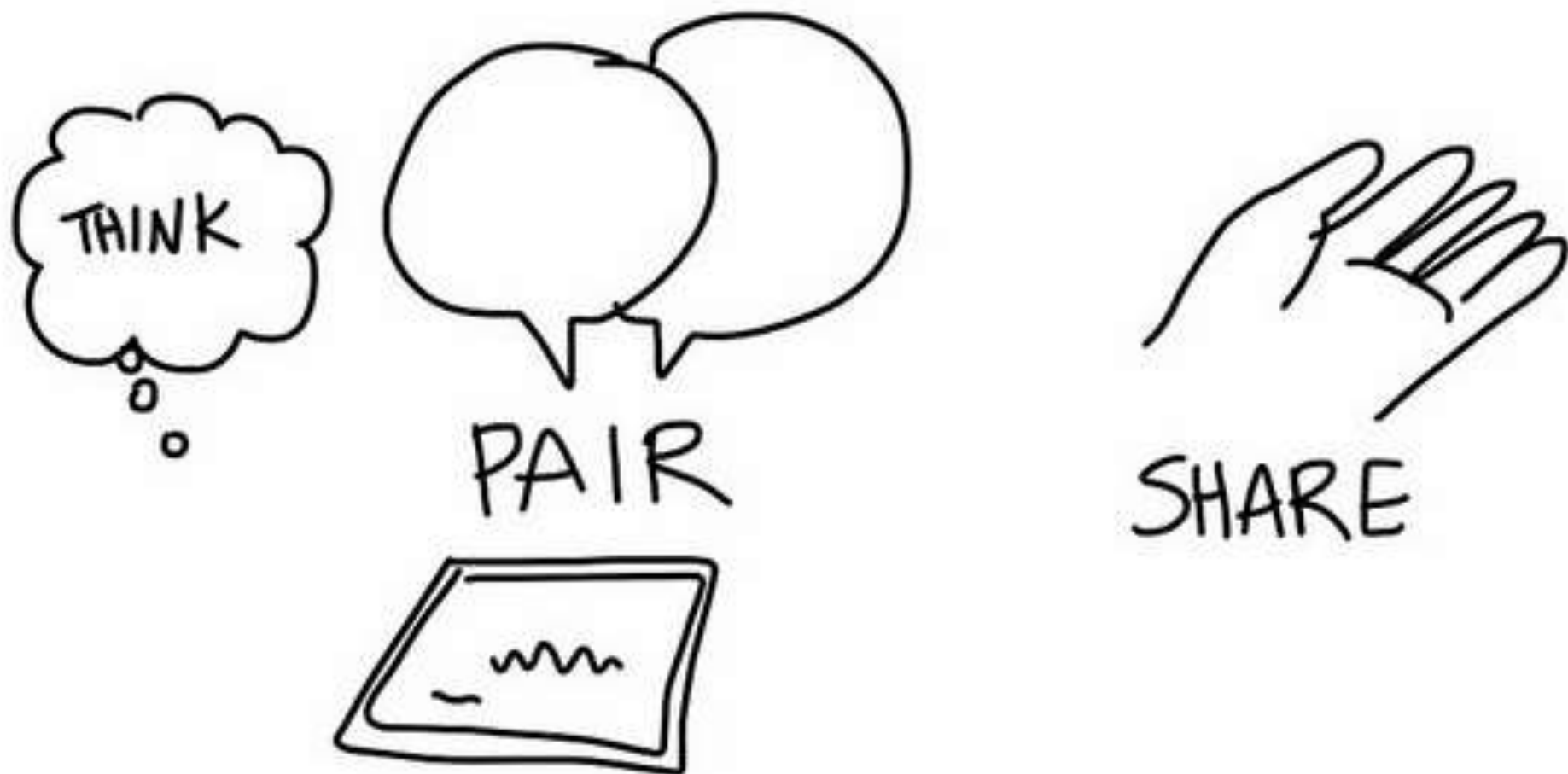


Amdahl's Law

- Gene Amdahl [AMDA67]
- Potential **speed up** of program using **multiple processors**
- Concluded that:
 - Code needs to be parallelizable
 - **Speed up is bound**, giving diminishing returns for more processors

Amdahl's Law

- Task dependent, e.g.:
 - Servers gain by maintaining multiple connections on multiple processors
 - Databases can be split into parallel tasks



- A dual core processor is not necessarily faster than a single core processor
 - True or false?
 - Why?

Internet Resources

- <http://www.intel.com/>
 - Search for the Intel Museum
- <http://www.ibm.com>
- <http://gordonbell.azurewebsites.net/digital/timeline/tmlnhome.htm>
- Charles Babbage Institute
- PowerPC
- Intel Developer Zone

Internet Resources

Websites to look for

- WWW Computer Architecture Home Page
- CPU Info Center
- Processor Emporium
- ACM Special Interest Group on Computer Architecture
- IEEE Technical Committee on Computer Architecture
- Intel Technology Journal
- Manufacturer's sites
 - Intel, IBM, etc.

Reflection...

From the lecture today,
I learnt...

Link on eLearn / Chat